

— Abdominal walls

Objectives

- Anterolateral abdominal wall
 - Posterior abdominal wall
 - The diaphragm
 - The lumbar plexus
 - The aorta
-
- The inferior vena cava

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The posterior abdominal wall

1. Recognize different **muscles** of posterior abdominal wall (origin, insertion, nerve supply & action).
2. Describe the **diaphragm** (origin, insertion, nerve supply, actions & important opening).
3. Describe the **lumbar plexus** (roots, formation & branches)
4. Describe the **abdominal aorta** (beginning, termination & branches)
5. Describe the **inferior vena cava** (beginning, termination & tributaries)

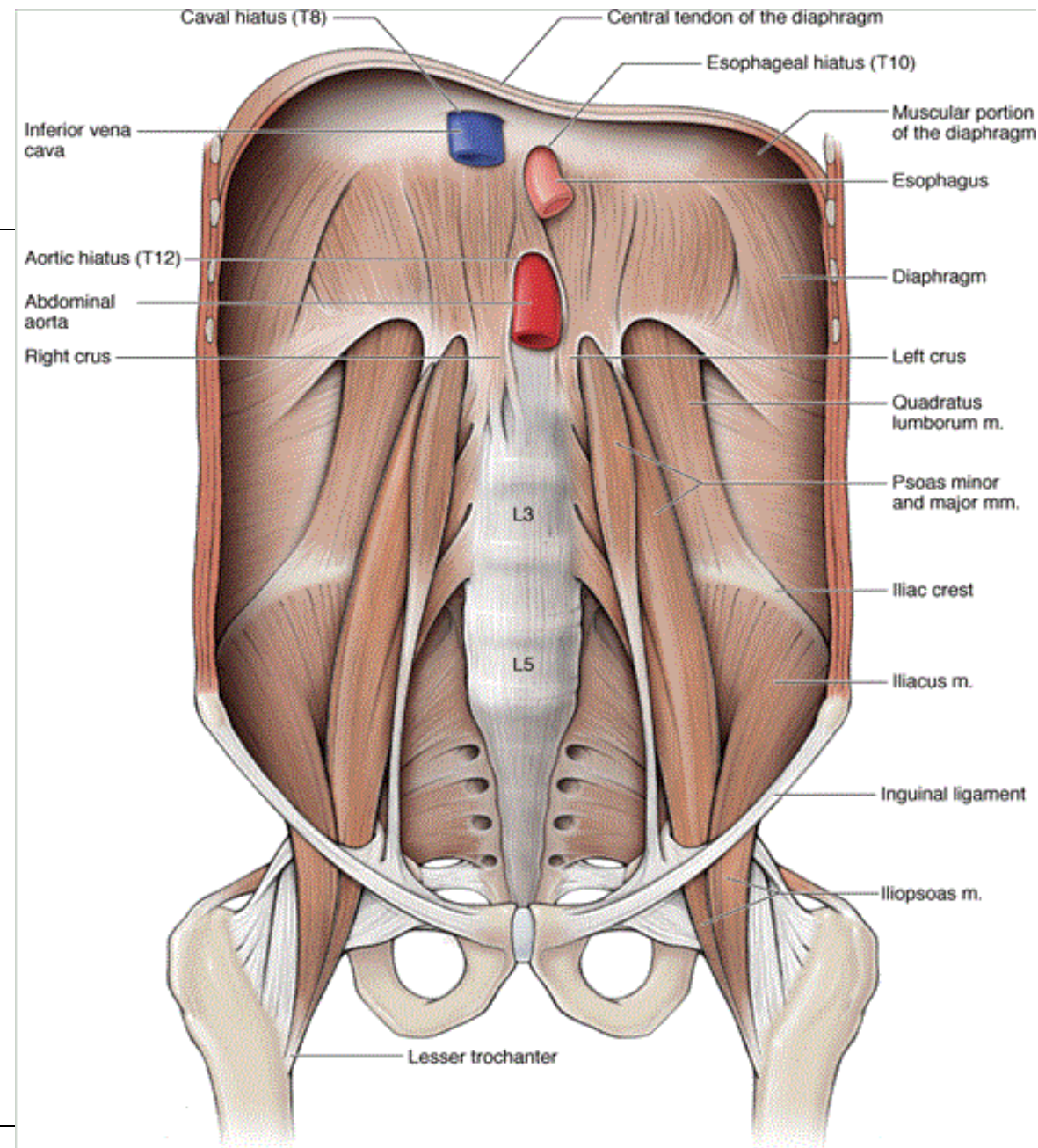
The posterior abdominal wall

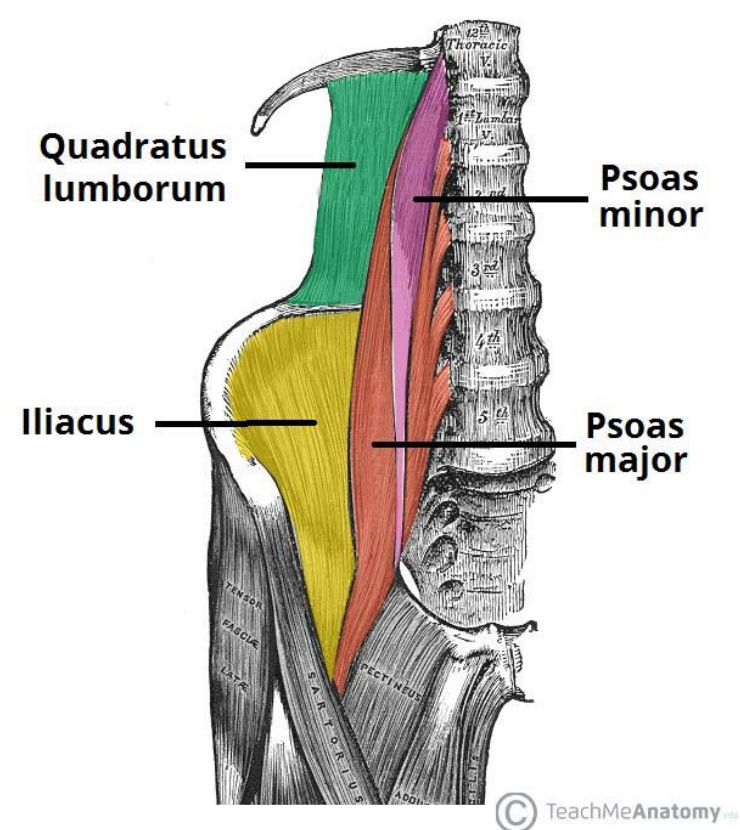
Muscles of posterior abdominal wall:

- 1- Psoas major muscle
- 2- Iliacus muscle
- 3- Quadratus lumborum muscle
- 4- Posterior part of transversus abdominus muscle
- 5- Crura of diaphragm
 - RT crus attached to L 1,2, & 3 vertebrae
 - LT crus attached to L 1 & 2 vertebrae

The **middle part** of posterior abdominal wall ..

It is formed by **5 lumbar vertebrae** & **intervertebral discs**,
They are covered by **anterior longitudinal ligament**





1- Psoas major muscle:

Origin : from transverse processes, vertebral bodies & intervertebral discs from **T12-L5**

Insertion ⑦ to lesser trochanter of femur

Nerve supply: from roots of **lumbar plexus**.

2- Iliacus muscle:

- **Origin**: iliac fossa of ilium.

- **Insertion** ⑦ to lesser trochanter of femur

Nerve supply: from **femoral nerve**

Actions of psoas major & iliacus (iliopsoas) :

A) Flexion of the thigh at hip joint

B) If thighs are fixed: they flex the trunk on the thighs (as in sitting up from lying position)

3- Quadratus lumborum muscle:

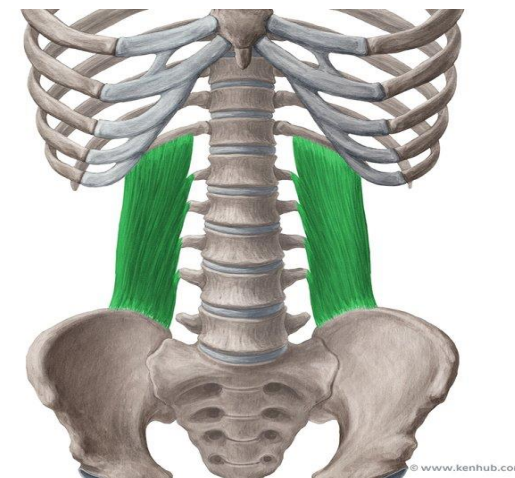
Origin → from iliac crest, iliolumbar ligament, & tips Of transverse processes of lumbar vertebrae

Insertion → into lower border of last rib (12th rib)

Nerve supply: roots of **lumbar plexus**

Actions:

- A) Fixation of last rib (during forced expiration),
- B) The muscles of one side → Lateral flexion of the trunk,
- C) The muscles of both sides → **extend the trunk**.



The Diaphragm

Origin inner surface of thoracic outlet:

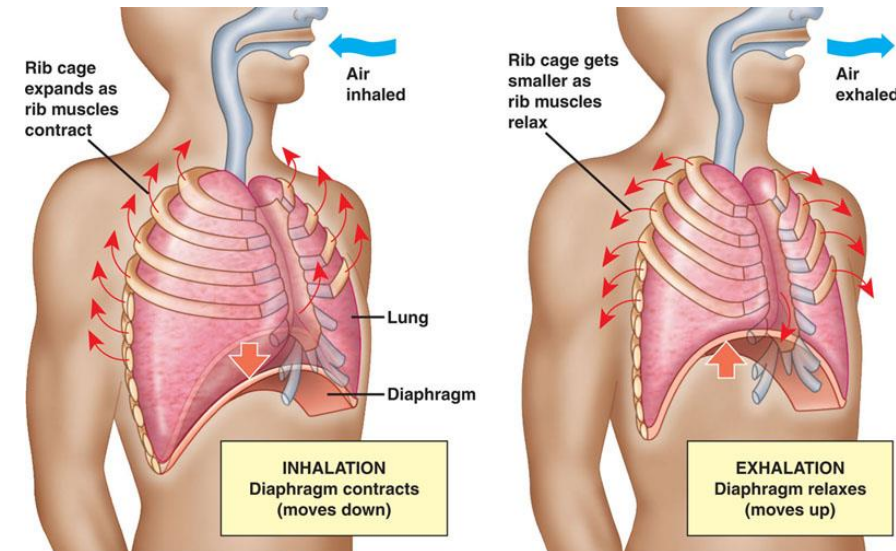
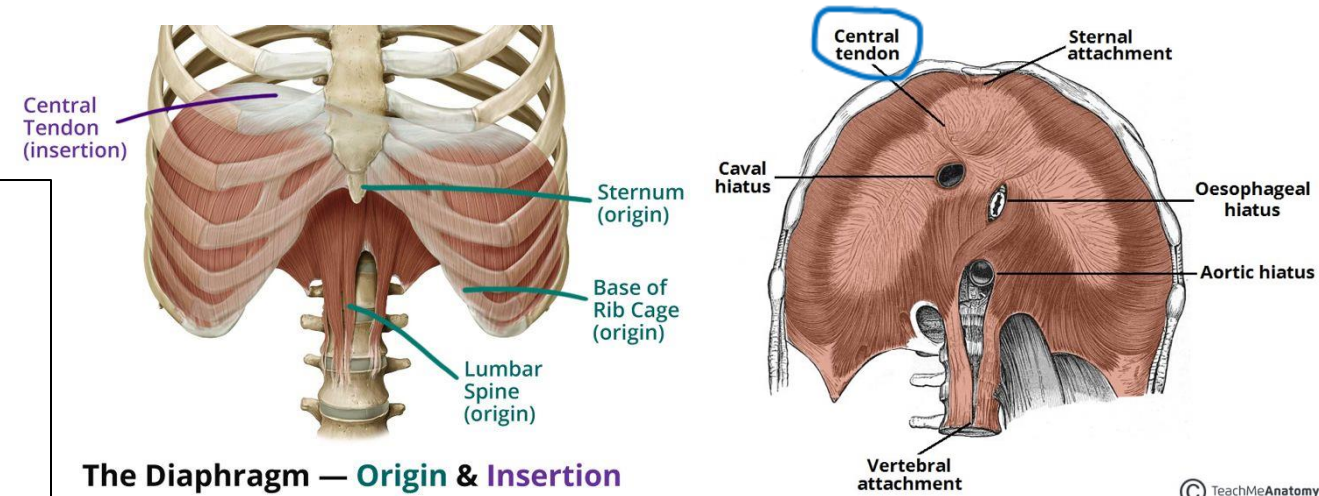
- 1- **Sternal origin** → from **xiphoid process**
- 2- **Costal origin** → from **lower 6 costal cartilages**
- 3- **Vertebral origin** → **right** crus of diaphragm: from **L1, 2, 3.**
left crus of diaphragm: from **L1, 2 only**

Insertion

Fibers converge and directed centrally toward the **Central tendon of the diaphragm**.

Actions / Functions of the diaphragm:

- 1- It is the chief muscle of **respiration**
- 2- Muscle of **abdominal straining** (increase intra- abdominal pressure), aids muscles of the anterior abdominal wall ⑦ to evacuate abdominal or pelvic contents e.g. **defecation, labor, urination, vomiting,...**
- 3- **Thoracoabdominal pump:** helps in venous & lymph return: by descend of diaphragm ⑦ **positive intra-abdominal pressure & negative intrathoracic pressure**



Major openings of the diaphragm:

Caval opening: at level of **T 8**, it transmits:

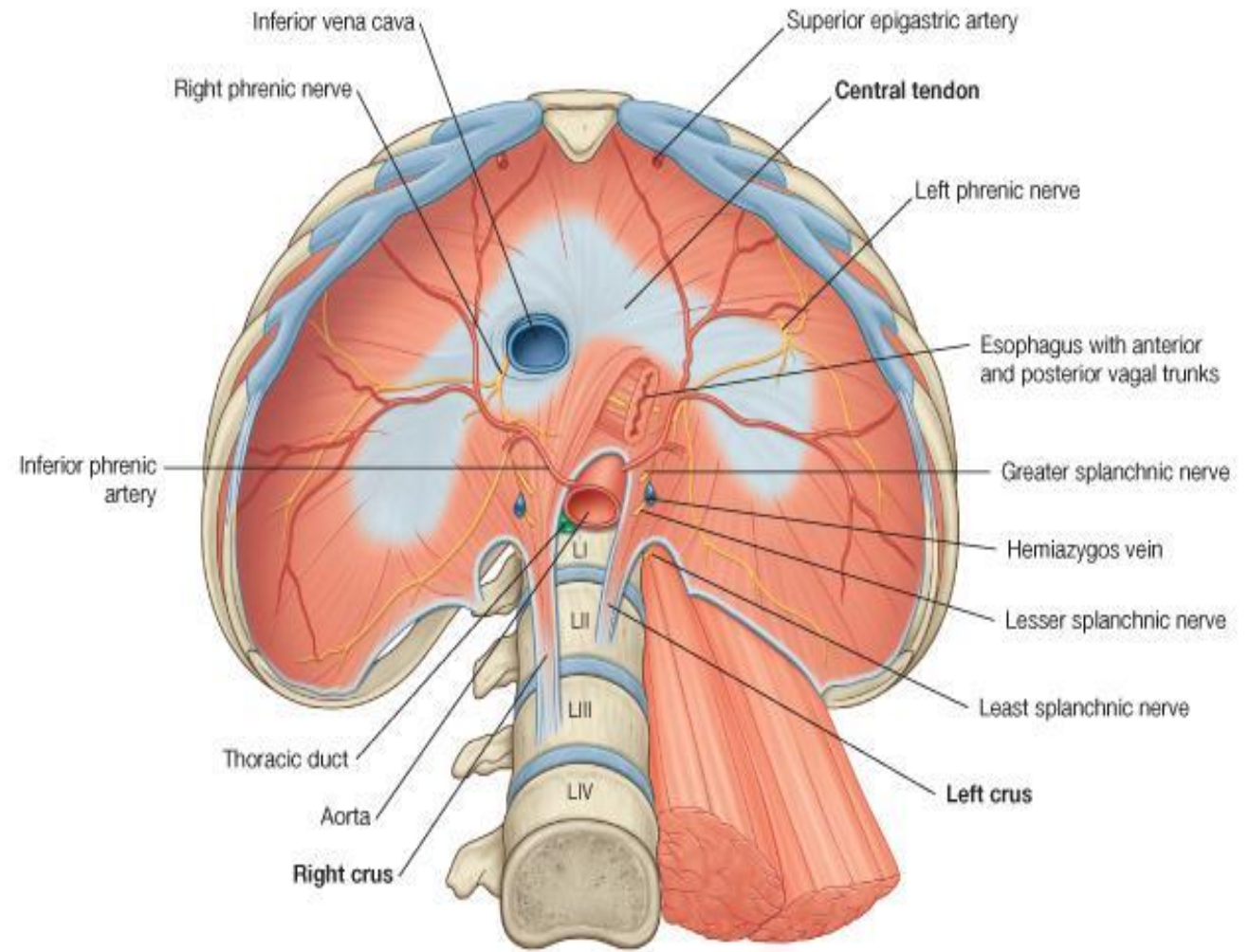
- 1- IVC,
- 2- RT phrenic nerve,
- 3- Lymph vessels

Esophageal opening: at level of **T 10**, it transmits:

- 1- Esophagus
- 2- LT & RT vagus nerves.
- 3- LT gastric blood vessels.

Aortic opening: at level of **T 12**, it transmits:

- 1- Aorta
- 2- Thoracic duct
- 3- Azygos vein



Lumbosacral plexus

The lumbosacral plexus is a **network of nerves derived from lumbar and sacral roots**. It is formed by **ventral rami of the lumbar and sacral nerves, T12- L1 to S4**.

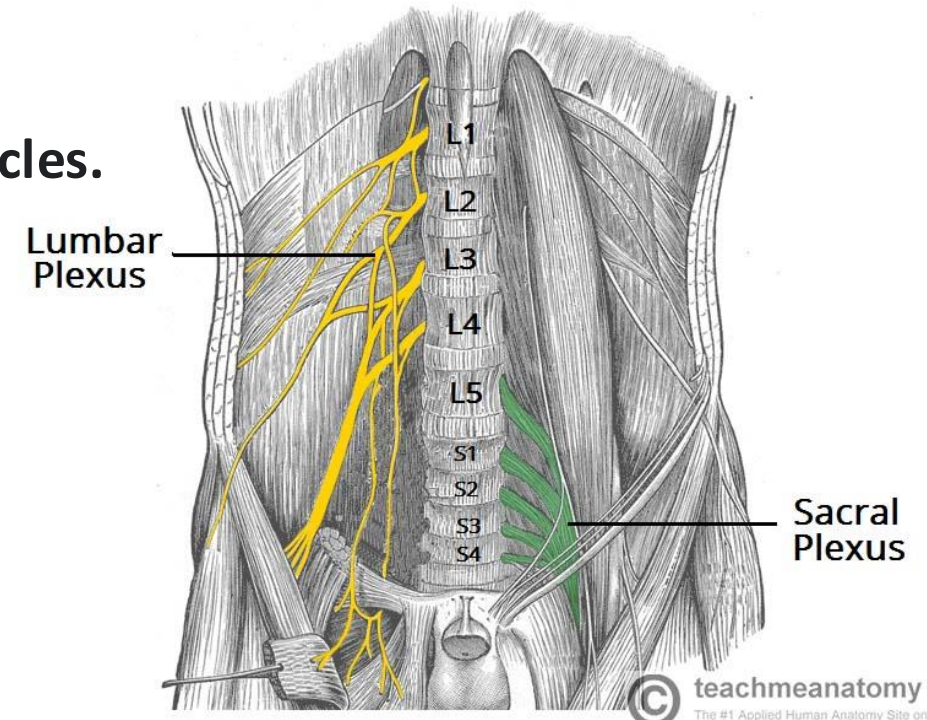
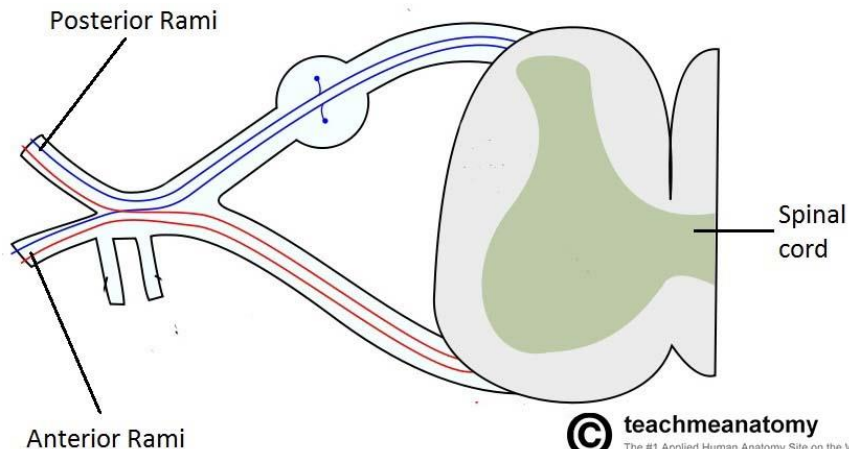
The **lumbar part** is formed by roots from **L1-L4** roots.

The **sacral part** is formed by roots from **L5–S4** roots.

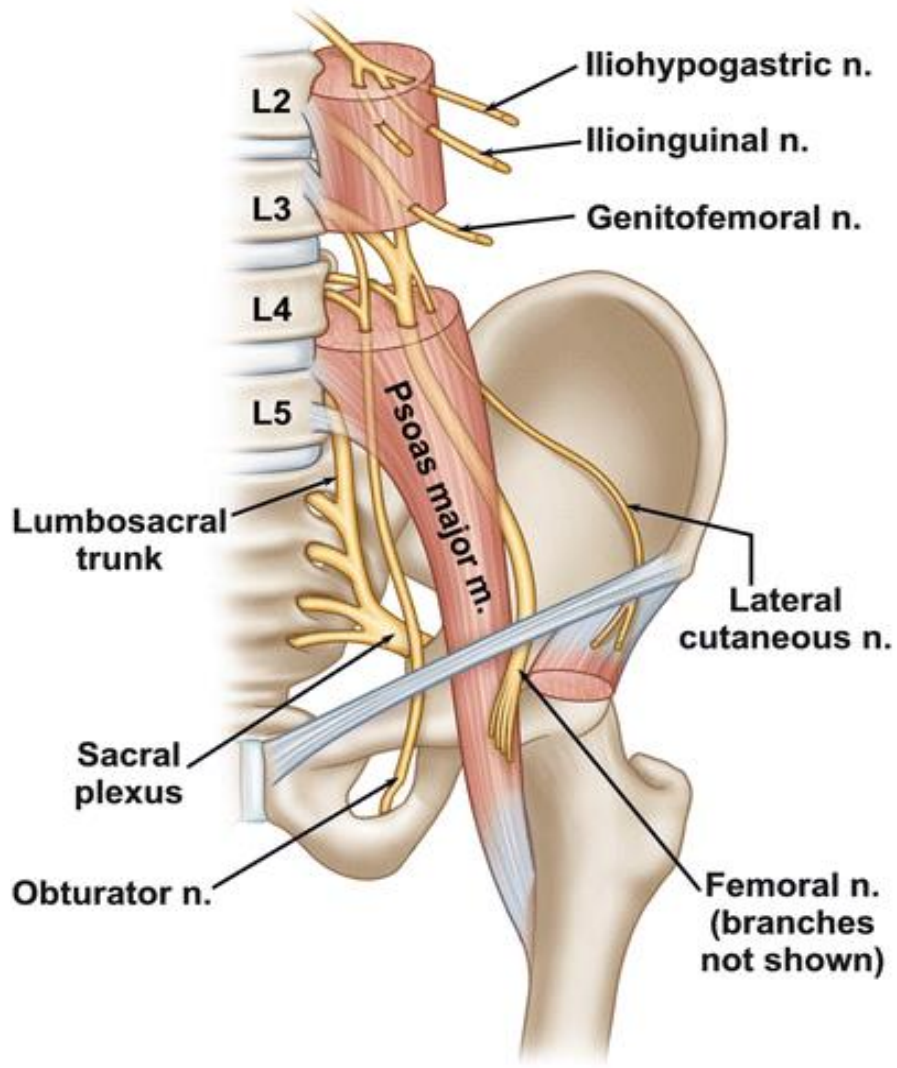
These divide into anterior & posterior divisions, which give rise to anterior & posterior branches, respectively.

The **anterior** branches supply the **flexor muscles of thigh**.

The **posterior** branches supply the **extensor & abductor muscles**.



The lumbar plexus



- 1- It is formed by ventral rami of L 1,2,3, & 4
- 2- It is formed inside the **psoas major muscle**
- 3- Its branches arise from the psoas major

Branches arising at lateral border of psoas major:

- 1- Iliohypogastric nerve
- 2- Ilioinguinal nerve
- 3- Lateral cutaneous nerve of thigh
- 4- Femoral nerve

Branches arising at medial border of psoas major:

- 1- Obturator nerve
- 2- Lumbosacral trunk

Branches arising at front of psoas major:

Genitofemoral nerve

Branches of the lumbar plexus

The iliohypogastric nerve

It runs to the iliac crest, on abdominal wall.

Motor Functions: Innervates the **internal oblique** and **transversus abdominis**.

Sensory Functions: Innervates the skin of lower anterior abdominal wall and posterolateral gluteal region.

The ilioinguinal nerve

After innervating the muscles of the anterior abdominal wall,
It passes through the superficial inguinal ring.

Motor Functions: Innervates the **internal oblique** and **transversus abdominis**.

Sensory Functions: Innervates the skin of genitalia & upper antero-medial of thigh.

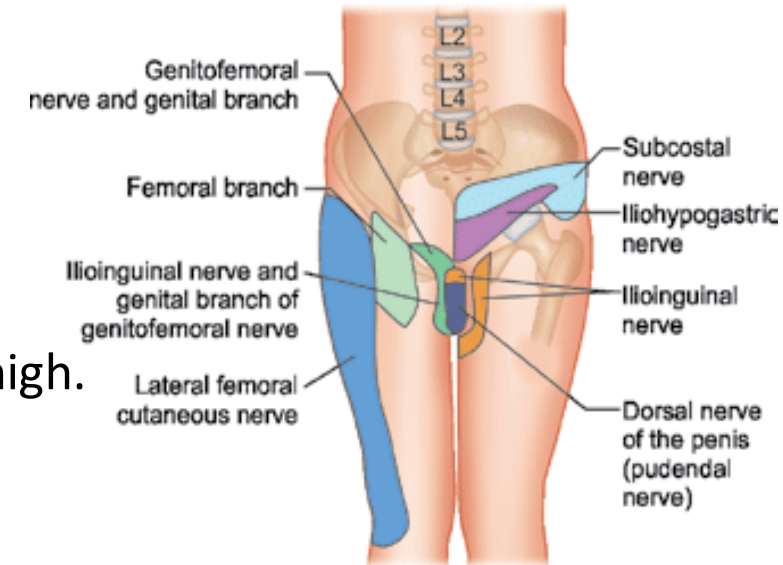
The genitofemoral nerve

After leaving 'psoas major muscle, it divides into a genital branch, and a femoral branch.

Motor Functions: The genital branch innervates the **cremasteric muscle**.

Sensory Functions: The genital branch innervates the **skin genitalia** (in males and females).

The femoral branch innervates the **skin on the upper anterior thigh**.



Branches of the lumbar plexus

The lateral Cutaneous Nerve of the Thigh

It enters the thigh at the lateral aspect of the inguinal ligament

This nerve is a pure **sensory**.

Sensory Functions: Innervates the lateral thigh down to the level of the knee.

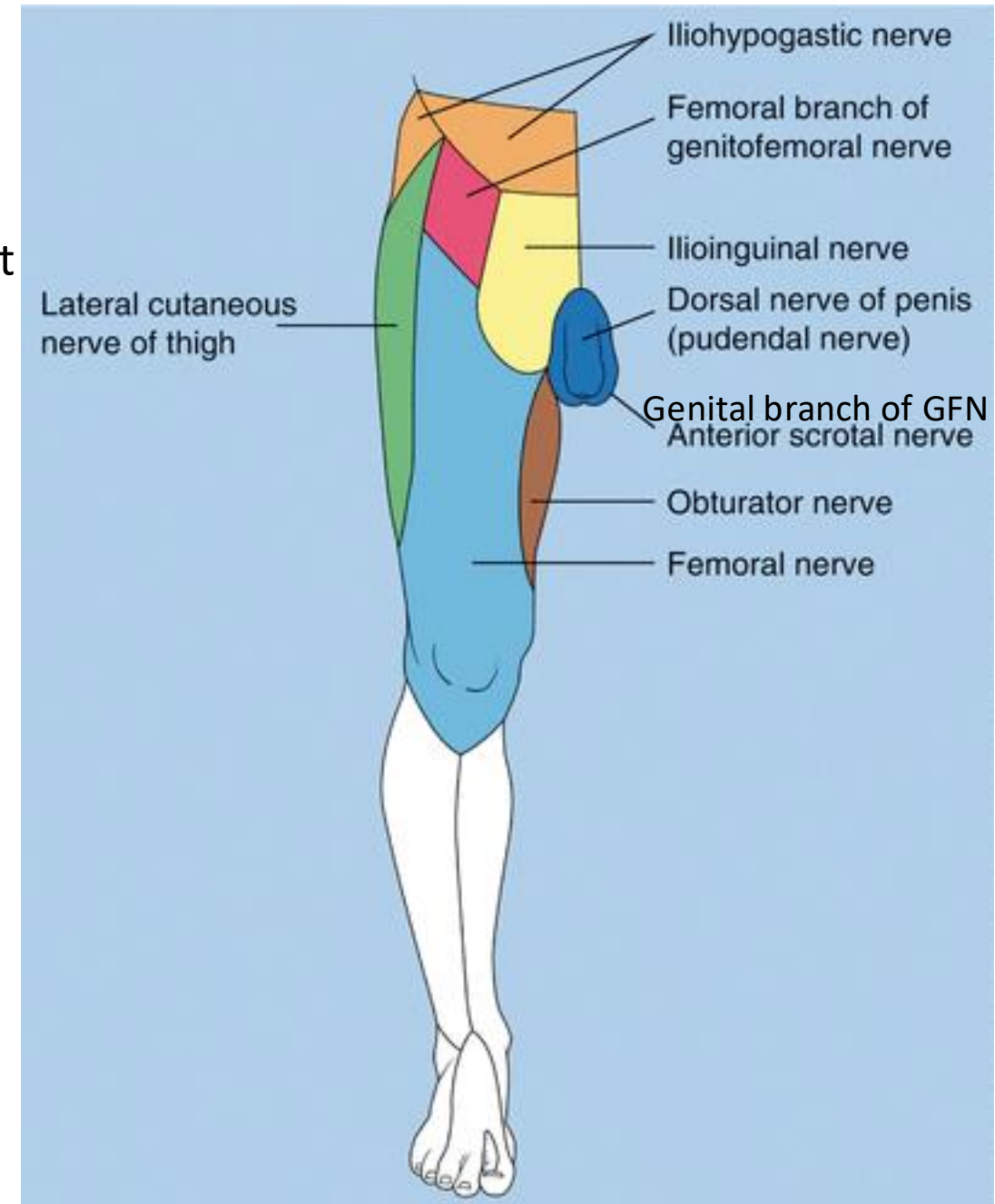
The obturator nerve

The femoral nerve

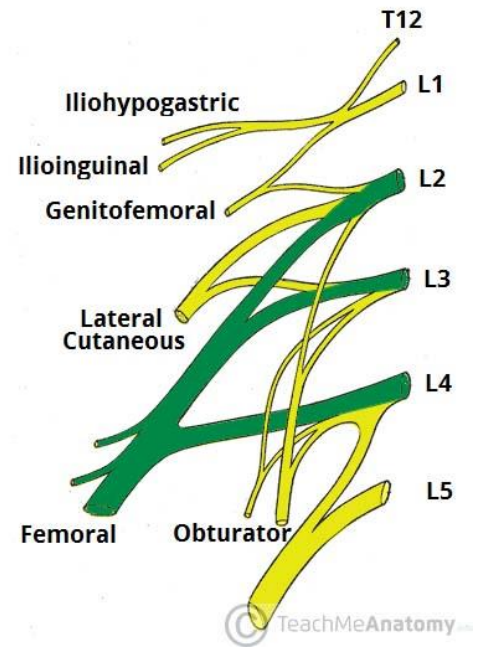
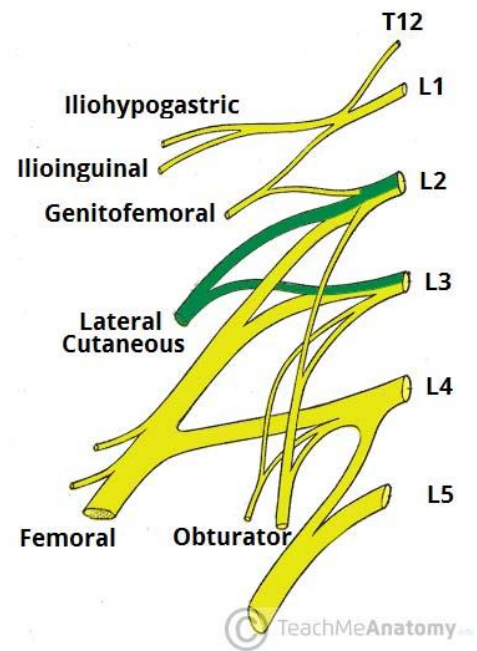
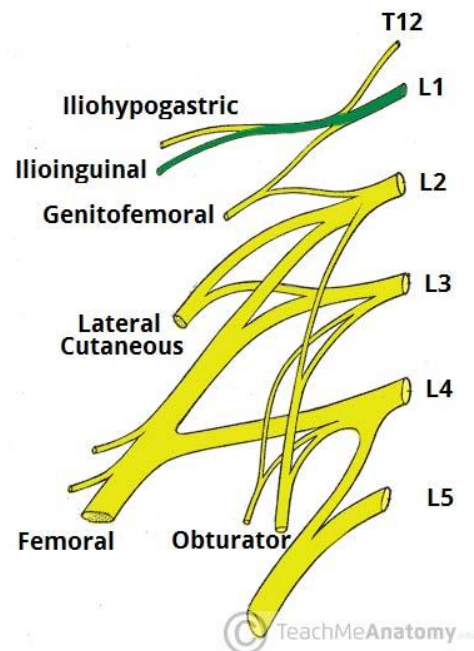
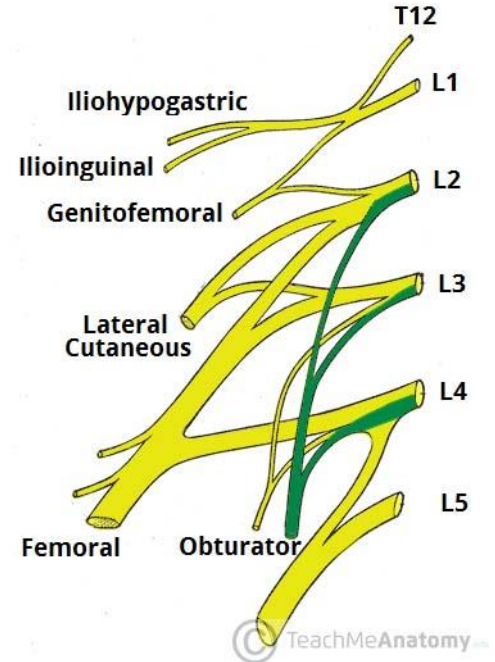
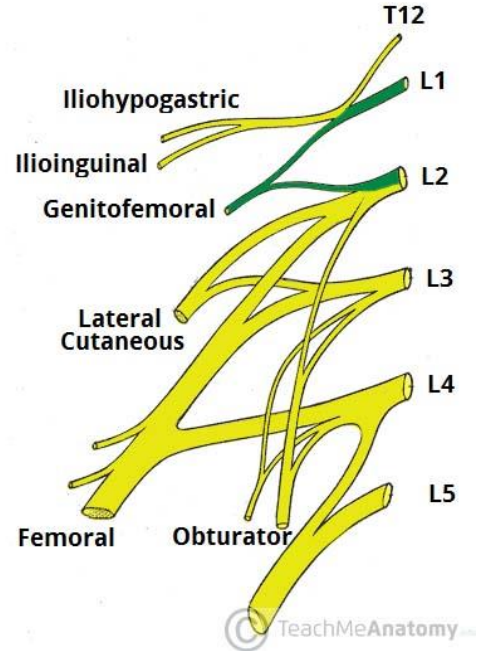
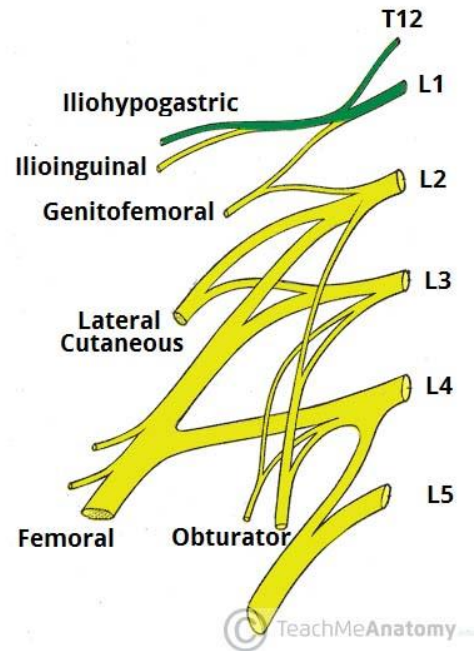
Name the branch ..

Give the root value..

Know its Motor & sensory branches.



Ventral rami from L1-L4



The Abdominal Aorta

- **Origin:** it begins at **T 12**, as a continuation of descending thoracic aorta.
- **Course:** it descends in front of lumbar vertebrae
- **Termination:** at **L 4** by dividing into RT & LT common iliac arteries

Branches of abdominal aorta:

A) Single branches:

- 1- Celiac artery
- 2- Superior mesenteric artery (SMA).
- 3- Inferior mesenteric artery (IMA)
- 4- Median sacral artery

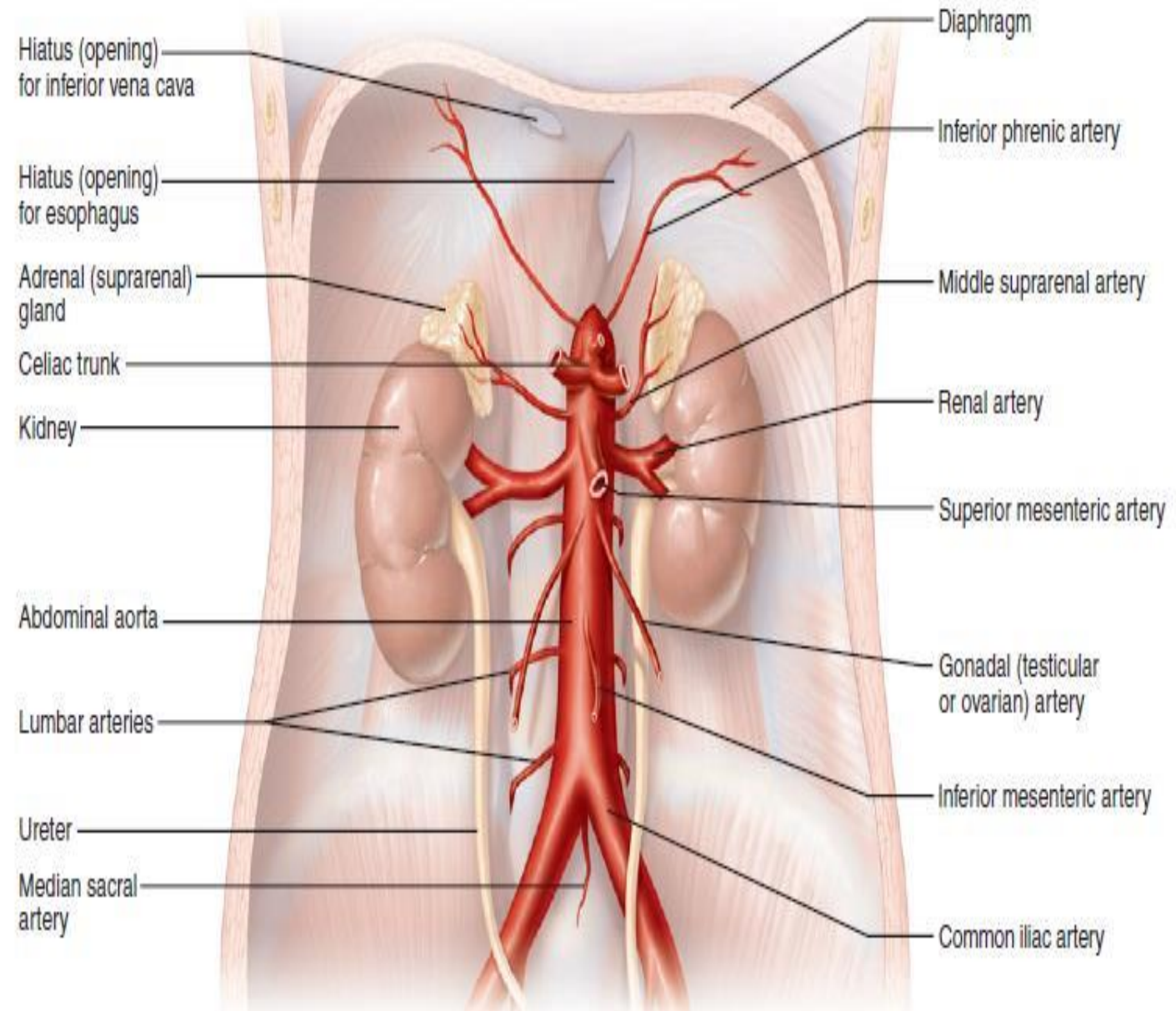
B) Paired branches: (RT & LT)

- 1- Inferior phrenic arteries
- 2- Suprarenal arteries
- 3- Renal arteries
- 4- Gonadal arteries (testicular or ovarian)

C) Lumbar: Four lumbar arteries

D) Terminal branches

Common iliac arteries (RT & LT)



The Inferior vena cava

It conveys venous blood from body below diaphragm ⑦ to RT atrium

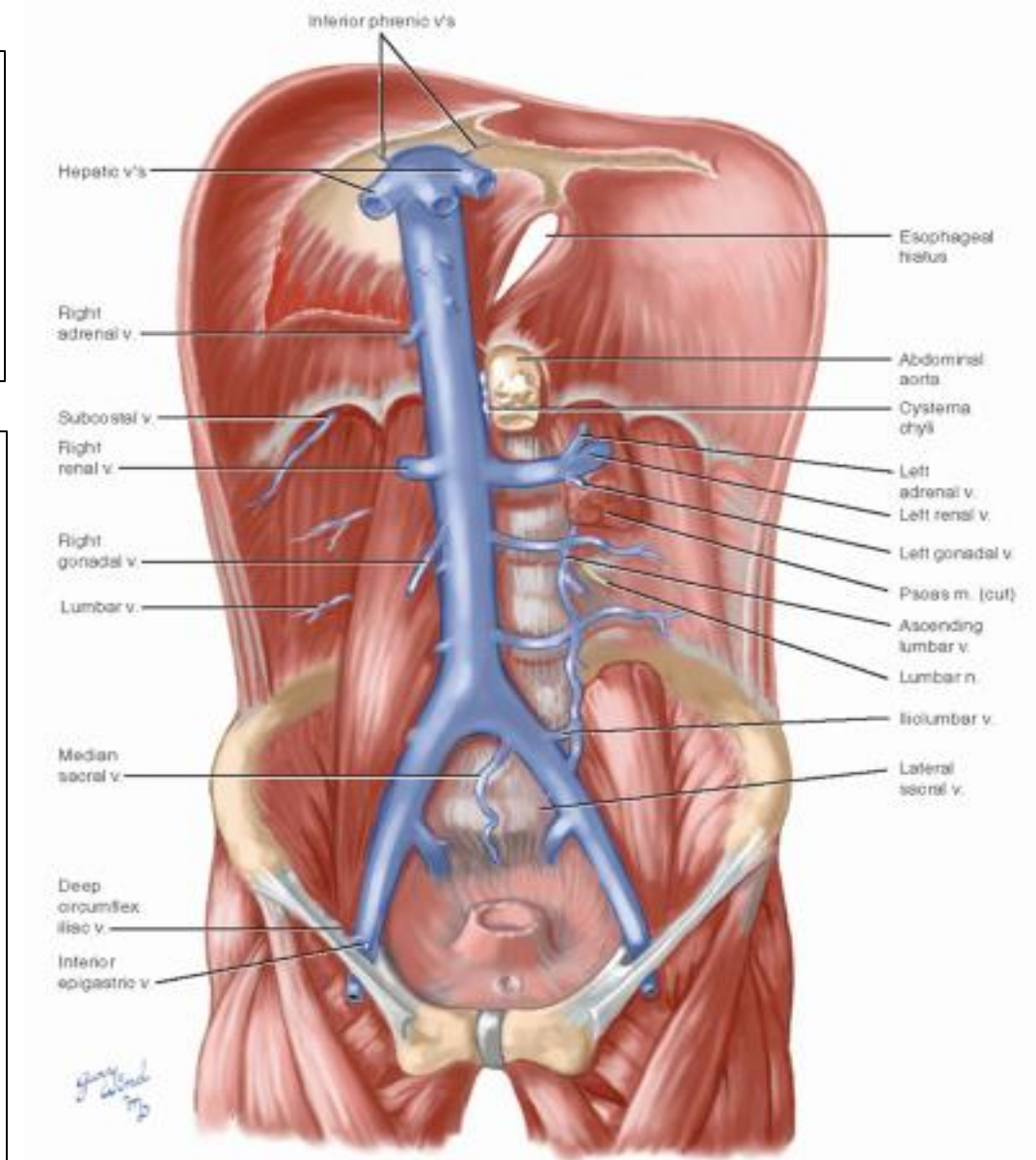
- **Origin** formed by union of 2 common iliac veins at level of **L5**
- **Course**: it ascends to the **RT side of abdominal aorta**
- **Termination**: in the RT atrium of the heart

Tributaries of IVC :

- 1- Common iliac veins (2 veins of origin)
- 2- lumbar veins (paired)
- 3- inferior phrenic veins (paired)
- 4- Renal veins (paired)
- 5- **RT** gonadal vein
- 6- **RT** Suprarenal vein
- 7- Hepatic veins (2 – 3 veins)

NB:

- **Median sacral vein** ⑦ ends in the **LT** common iliac vein
- **LT gonadal vein & LT suprarenal vein** ⑦ end in the **LT** renal vein

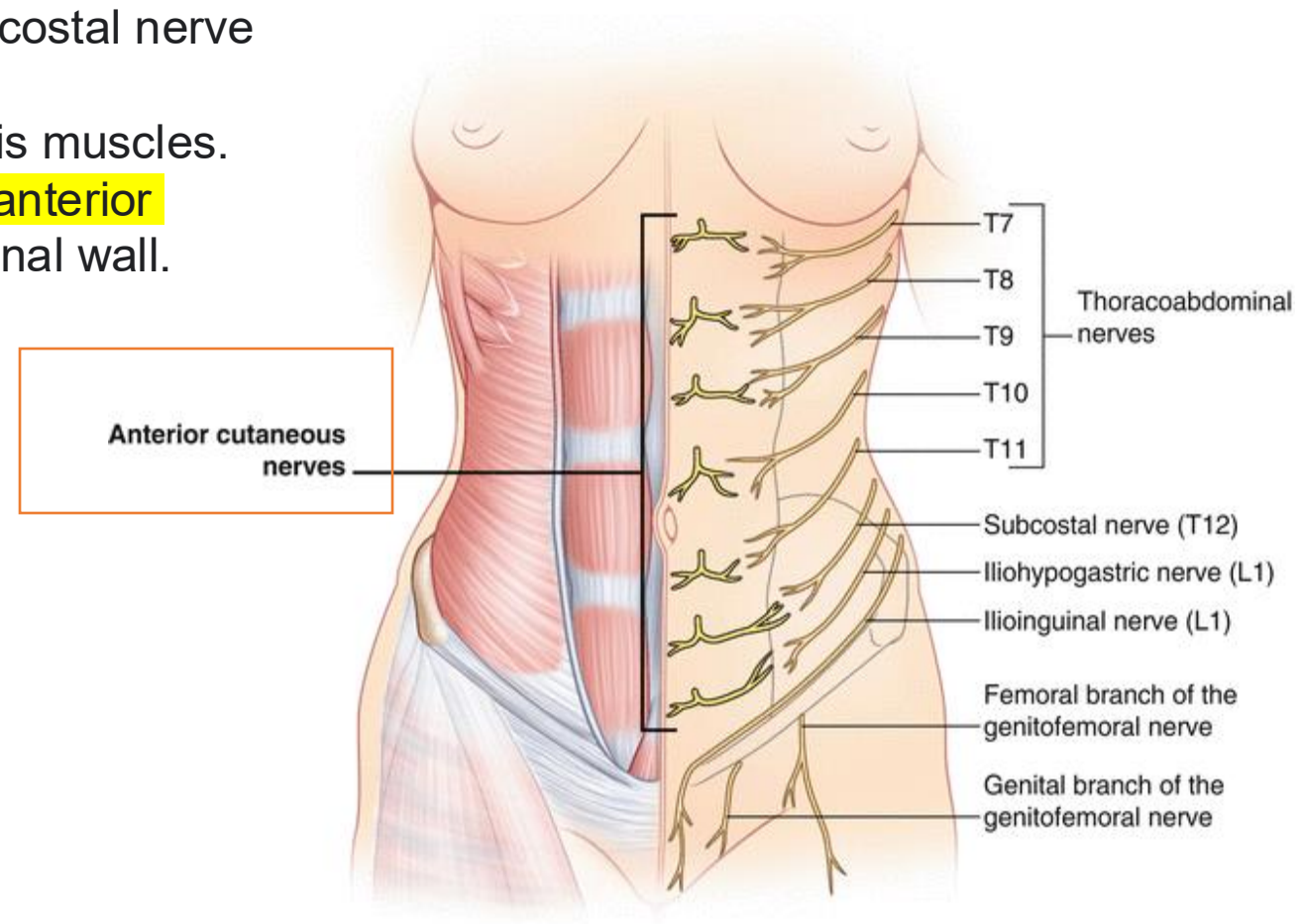


Cutaneous innervation of anterior abdominal wall:

At the mid-axillary line, the intercostal nerves and subcostal nerve (T7-T12) give off **lateral cutaneous branches**. These nerves run along internal oblique & transversalis muscles. Then, they enter subcutaneous tissue to become the **anterior cutaneous branches** of the skin in the anterior abdominal wall.

with **T7 and T8** predominantly supplying sensation to the inferior lateral thoracic wall

and **T9 to T12** predominantly supplying sensation to the lateral abdominal wall.



Lymphatic drainage of anterior abdominal wall:

A) Superficial lymphatics:(follow veins):

1-Above the umbilicus →into **anterior** (pectoral) group of axillary lymph nodes

2-Below umbilicus →into **superficial** inguinal LN

B) Deep lymphatics:(follow arteries):

1-Above the umbilicus →to internal thoracic (parasternal) LN

2-Below the umbilicus →to **external** iliac LN

3-Lateral parts →into para-aortic (lumbar) LN &
posterior mediastinal LN

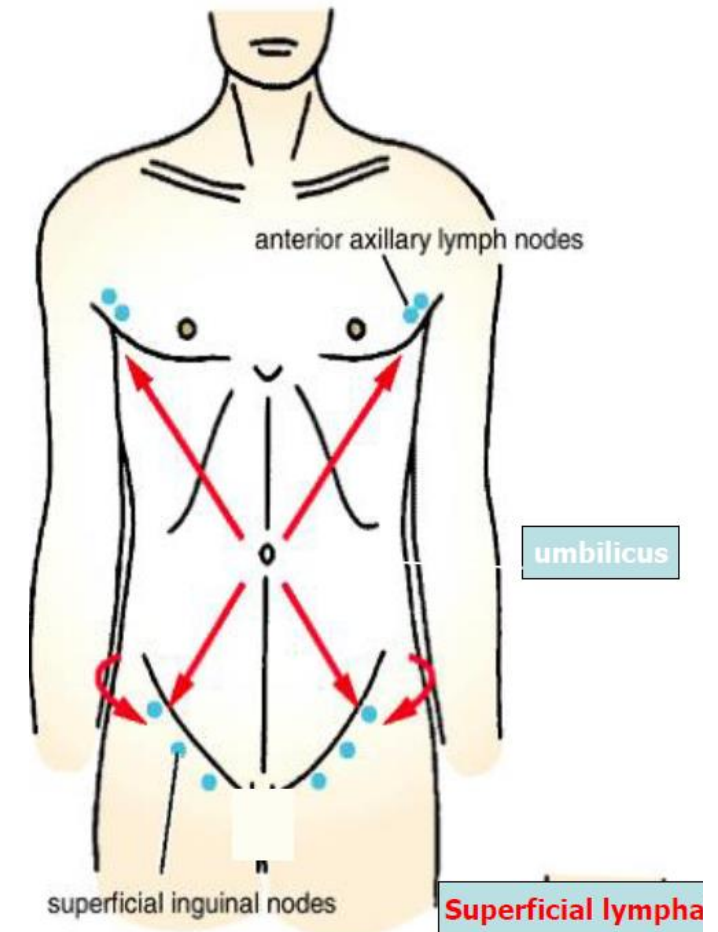
(along lumbar & posterior intercostal arteries)

NB:

-Lymphatic drainage of the **back of the trunk (superficial):**

1-Above iliac crest →into posterior (subscapular) group of axillary LN

2-Below iliac crest →into inguinal LN



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Thank you